

District Cooling System: Toward a Sustainable Future

AESL provides DCS solution as a “cooling-as-a-service” model, where the entire capex is borne by the company, users pay solely based on their usage

As we brace ourselves to tackle the challenges of climate change and meet the demands of ever-increasing cooling, district cooling system (DCS) emerges as a promising solution. As opposed to conventional cooling system, the objective behind DCS is to achieve energy efficiency and sustainability, simultaneously reducing operational costs and environmental impact.



What is District Cooling System?

A district cooling system or DCS is an energy and cost-efficient cooling system, which is designed to provide cooling for air conditioning in large-scale urban areas, commercial complexes, and industrial facilities. It's a centralised cooling solution from where cooling is distributed to multiple buildings through a network of underground pipes. This centralised nature of DCS allows for economies of scale, resulting in reduced operational and maintenance costs.

Additionally, the environmental benefits of district cooling systems are also profound. By reducing energy consumption and peak demand, DCS help in lowering greenhouse gas emissions. The integration of renewable energy sources, such as solar and wind power, into the cooling network further diminishes the carbon footprint of cooling operations.

Toward Sustainable Future

DCS and other innovative centralised cooling solutions not only have the potential to reduce power consumption by almost 35% and peak demand by 30%, but also contribute to reduction in power supply infrastructure cost by as much as 45%.

Recognising this synergy, Adani Energy Solutions Limited (AESL) entered this business which will not just optimise its infrastructure costs, power procurement and transmission costs but also help in demand side management. Further, as a power transmitter and distributor, Adani can ensure the blending of green power